

Evaluating Agentic AI for Supply-Chain Disruption Mitigation

A Two-Phase Paired-Simulation Study of a Bounded LLM Agent Against a Classical Heuristic

Extended Edition – with full-grid visuals and a reinforcement-learning follow-up

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Abstract

Can a bounded, tool-using large language model (LLM) agent mitigate supply-chain disruption more effectively than a transparent, cost-minimizing classical heuristic – and if so, under what conditions? We answer this in two phases, sharing one paired-simulation infrastructure throughout. **Phase 1** tested one network topology against three disruption severities (300 real, live-API replications) and found the LLM agent losing under Light disruption but winning decisively under Medium/Heavy – with every one of 300 replications agreeing unanimously (100%/0% per profile), a result too clean to trust at face value. Auditing why traced the unanimity to insufficient environmental randomness and a single, low-redundancy topology, not to a genuine effect ceiling. **Phase 2** redesigned the experiment along exactly the axes the audit implicated: three network topologies crossed with the same three severities (nine cells, 498 replications), plus per-replication randomness in disruption timing, duration, disclosure, and shipment volume. The redesign both validated Phase 1 (the original topology reproduces its original pattern almost exactly) and overturned its implicit generality: **network topology does not merely scale the LLM’s advantage – it can reverse its sign**, with the highest-redundancy topology favoring the heuristic ever more strongly as severity rises, the opposite of the other two topologies. Across the whole two-phase program, agentic AI shows a real, statistically confirmed advantage for high-stakes disruption response on constrained networks, and a real, equally confirmed disadvantage under low-stakes conditions and high-redundancy networks – a nuanced, conditional answer, not a universal one. This edition adds the full-grid heatmap and the signal-to-noise validation figure, plus a follow-up section on a reinforcement-learning formulation of the same decision problem.

1 Research Problem

Agentic AI – LLM-based systems that reason, call tools, and act – is increasingly proposed for operational supply-chain decisions: rerouting, expediting, or cancelling shipments in response to disruption, where a wrong call has directly measurable cost. The claim that “AI beats the baseline” rarely survives scrutiny in this domain: mismatched scenarios, cherry-picked examples, or a classical baseline never given a fair, symmetric shot. This study’s overarching goal, spanning both phases reported here, is to test that claim rigorously for one narrow, well-defined decision – *what should happen to a shipment sit-*

ting at a node when the surrounding network is disrupted – and to answer not just *whether* an agentic policy helps, but *when*: **under identical disrupted conditions and identical operational constraints, can a bounded LLM policy reduce the incremental cost of disruption more effectively and more robustly than a transparent classical heuristic, and how does that advantage depend on network structure and disruption severity?**

Fairness is operationalized structurally, not asserted: both policies observe identical read-only state, are validated and fall back through the same machinery, and face the same stochastic demand, delay, and disruption realization within a replication. Each policy is scored against its own undisrupted counterfactual, isolating the *incremental* cost of disruption rather than baseline operating cost.

2 Related Work

LLM agents with tools. ReAct [1] established the now-standard interleaved reasoning–action–observation loop for LLM agents, later made reliable via structured function calling; Toolformer [2] showed LLMs can learn *when* to invoke tools rather than merely being prompted to. Our agent follows this lineage: five deterministic, read-only tools, an 8-call budget, temperature 0, and a hard fallback to the classical heuristic on abstention or invalid output.

LLM agents in supply chain management. This is an active, recent direction: InvAgent [3] applies multi-agent LLMs to inventory management under demand uncertainty; broader surveys [4] catalog knowledge-integration and negotiation as dominant use cases; [5] studies autonomous multi-agent consensus across supply-chain roles. This literature consistently flags a verification gap – agentic LLMs can inherit biases and produce confidently wrong decisions absent explicit checking – which is precisely why this study reports null and unfavorable results for the agent alongside favorable ones, rather than only the latter.

Classical disruption management. The heuristic baseline sits in a long tradition of cost-minimizing routing and recovery rules under disruption [6, 7]: transparent, auditable, structurally incapable of considering information outside its candidate-route cost estimate. It is not a strawman – it wins cleanly in both phases under Light disruption, and dominates one topology entirely in Phase 2.

Severity	Mean Δ	95% CI	LLM win
Light	+992	[967, 1018]	0%
Medium	-3,911	[-4011, -3811]	100%
Heavy	-74,490	[-75919, -73061]	100%

Table 1: Phase 1: mean Δ and 95% CI, $n = 100$ /severity. Every one of the 300 replications agreed unanimously with every other in its profile.

3 Shared Methodology

Both phases run on one unmodified simulator: a discrete-time, single-product network (suppliers $\rightarrow$ ports $\rightarrow$ hubs $\rightarrow$ plant) in which a policy intervenes on at most one shipment per decision point with exactly one action – WAIT, REROUTE, EXPEDITE, or ABSTAIN – validated by shared rules with a heuristic-then-WAIT fallback on failure.

One random seed drives one *replication*: demand, ordinary transport delay, and (Phase 2 only) disruption realization and shipment volume are drawn entirely before either policy acts. Four cloned branches then run per replication – {heuristic, LLM} $\times$ {undisrupted, disrupted} – sharing an identical 20-day warm-up snapshot. Per policy p , $TCD_p = J_p^{\text{disrupted}} - J_p^{\text{undisrupted}}$ (J sums transport, reroute, expedite, holding, backlog, late, and terminal costs); the reported effect is $\Delta = TCD_{\text{LLM}} - TCD_{\text{heuristic}}$ ($\Delta < 0$ favors the LLM). Pairing removes common-mode noise, since the same random world hits both policies, giving far more power per replication than an unpaired design.

4 Phase 1: Initial Evaluation

Phase 1 tested one network (5 nodes, 6 edges, ~ 3 end-to-end routes) against three fixed disruption severities – a partial-capacity cut (Light), a full 7-day port closure (Medium), and a 3-week closure plus alternate-route congestion (Heavy) – 100 live-API replications each, 300 total.

Headline result. The LLM agent loses under Light disruption but wins decisively under Medium and Heavy (Table 1) – a clean, seemingly interpretable severity gradient. **The problem:** within every profile, all 100 replications agreed with each other exactly (100%/0%), never once split. Auditing why traced this to three compounding, fixable design properties, not a genuine effect ceiling: (i) the shipment release schedule was fully deterministic, exactly 40 units every day; (ii) the disruption’s timing, duration, and disclosure were fully known in advance; (iii) the network offered only ~ 3 routes – not enough structure for a close call to ever arise. Quantitatively, the mean $|\Delta|$ was $7.6 \times -10.2 \times$ larger than the between-replication standard deviation of Δ in every profile: the experiment could not generate a world where the effect and the noise were remotely comparable in scale, so unanimity was close to guaranteed by construction, not by an overwhelming true effect. This motivated Phase 2.

5 Phase 2: Topology-Generalized Evaluation

Phase 2 targets the audit’s three causes directly, without altering the shared simulator, action space, validator, cost model,

Topology	Severity	Mean Δ	95% CI	LLM win
Compact	Light	+935	[814, 1055]	0%
Compact	Medium	-1,436	[-2210, -663]	75.8%
Compact	Heavy	-56,915	[-68896, -44934]	100%
Standard	Light	+921	[857, 985]	0%
Standard	Medium	-4,192	[-4682, -3702]	100%
Standard	Heavy	-111,850	[-122960, -100741]	100%
Extended	Light	+1,636	[1451, 1820]	0%
Extended	Medium	+5,176	[4536, 5817]	0%
Extended	Heavy	+22,114	[17950, 26279]	0%

Table 2: Phase 2: mean Δ and 95% CI on all nine cells. $n = 33$ (Compact/Extended), $n = 100$ (Standard). Every CI excludes zero.

or pairing logic (Section 3). **Topology:** three tiers – Compact (4 nodes/4 edges/0 non-emergency reroutes), Standard (Phase 1’s original network, byte-identical), and Extended (10 nodes/16 edges/7 structurally distinct non-emergency routes, confirmed by betweenness centrality) – crossed independently with the same three severities. **Randomness:** disruption start day, duration, and disclosure delay are now sampled per replication rather than fixed (a zero-uncertainty template provably reproduces Phase 1 bit-for-bit); shipment quantity is sampled per release on Compact/Extended (Standard keeps Phase 1’s fixed 40 units by design, as an unmodified behavioral anchor). Extended’s severity scenarios target hub_1 rather than port_primary, since centrality analysis showed the latter is no longer that tier’s structural chokepoint once seven alternate routes exist.

Nine cells result, each an independently-run configuration. Compact and Extended run 33 replications/cell; Standard runs Phase 1’s original 100/cell as the reference column. All nine were calibrated (3 replications each) before the full run, which then executed live against the OpenAI API: 498 replications, $\sim 14,700$ LLM calls, ~ 4.5 h wall-clock across three concurrent per-topology processes, all nine completing without error.

Validation. Standard $\times$ Medium is Phase 1’s original configuration, re-run unmodified at Phase 2 scale: LLM wins 100% of 100 replications, matching Phase 1’s original pattern almost exactly (compare Table 1’s Medium row) – the shared infrastructure did not silently break. **Headline result:** Compact and Standard reproduce Phase 1’s shape, the LLM’s advantage growing with severity. **Extended inverts it** – the heuristic’s advantage grows with severity instead, winning every replication at every severity (Figure 1). Topology is not a magnitude dial on a fixed direction of effect; it can flip which policy is favored as conditions worsen. This is visible at a glance across the whole grid in Figure 2: no cell is hatched, since every 95% CI excludes zero, so every color shown is a real effect.

Separately, Compact $\times$ Medium becomes the first cell in either phase to land away from unanimity (75.8% LLM, 24.2% heuristic) – direct evidence the redesign produces genuine per-replication variation. Consistent with this, the signal-to-noise ratio $|\bar{\Delta}|/\text{sd}(\Delta)$ sits at $0.6 \times -3.0 \times$ across all nine Phase 2 cells, against the $7.6 \times -10.2 \times$ recomputed identically on Phase 1’s real runs (Figure 3) – the audited root cause is measurably fixed, not merely asserted fixed.

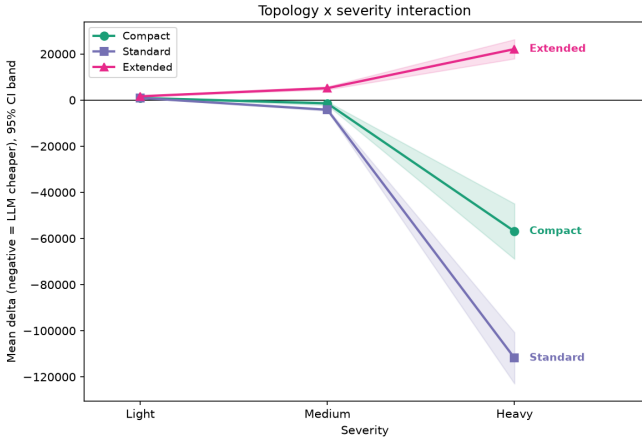
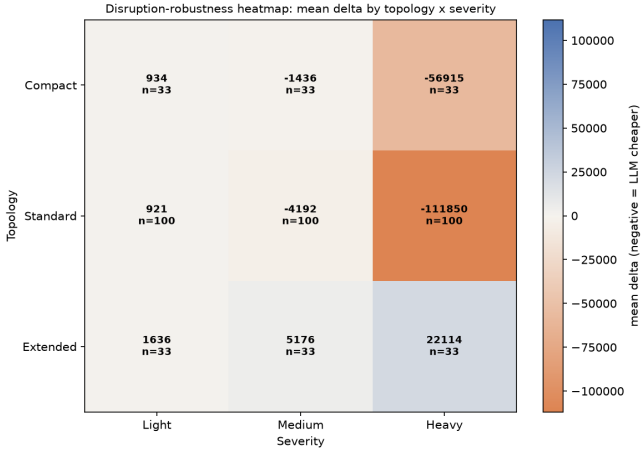


Figure 1: Phase 2: mean Δ vs. severity, one line per topology (signed-sqrt axis; bands are 95% CI). Compact/Standard diverge negative (LLM-favorable); Extended diverges positive (heuristic-favorable).



Hatched = 95% CI on mean delta includes zero (not distinguishable from a tie at this n). Blue = heuristic cheaper, orange = LLM cheaper. n<10 cells are calibration-stage, low-power estimates.

Figure 2: Phase 2: mean Δ across the full grid (diverging color; orange = LLM-favorable, blue = heuristic-favorable). Every cell’s effect is statistically real at its sample size.

6 Discussion: Synthesis Across Both Phases

Combining 798 replications across two phases and four network configurations, the study’s answer to its research question is conditional, not universal. **Pros of the agentic policy:** substantially larger cost reduction under severe disruption on constrained-to-moderate topologies (up to $-111,850$ mean Δ); the only policy to ever produce a genuinely mixed, non-unanimous outcome (Compact $\times$ Medium), suggesting situational judgment rather than a fixed rule; this advantage replicated independently across two separate live-API runs on the same network (Phase 1 and Phase 2’s Standard column). **Cons:** consistently worse under Light disruption in every configuration tested (WAIT is usually already optimal, and reasoning/tool-call overhead buys nothing); worse at every severity on the highest-redundancy topology; real monetary cost and ~ 2.6 s of live-API latency per decision, against a free, instant, deterministic heuristic evaluation. A plausible mechanism for the Extended reversal: the heuristic exhaust-

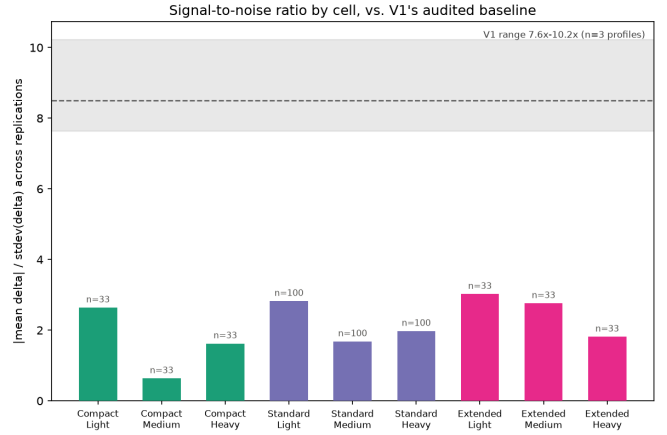


Figure 3: $|\bar{\Delta}|/\text{sd}(\Delta)$ per Phase 2 cell against Phase 1’s real, identically-computed baseline (shaded band). Every Phase 2 cell sits well under Phase 1’s $7.6\times-10.2\times$.

tively scores every candidate route by estimated cost and never has to judge which of many plausible options is worth pursuing, while a bounded agent (max 8 tool calls) reasons under genuinely higher ambiguity as the option count grows – plausible, not proven (see Limitations).

7 Limitations and Nuance

Four nuances qualify the combined findings. (1) Extended’s severity scenarios target hub_1, not port_primary – justified (Section 5), but it entangles the reversal with *which node* is attacked, not topology redundancy alone. (2) Standard’s shipment quantity stays fixed across both phases by design, while Compact/Extended draw real variance – an acknowledged confound on any comparison involving Standard specifically. (3) Phase 2’s 33 replications/cell is real, CI-confirmed power, but less than Phase 1’s 100/profile; Compact $\times$ Medium’s split could plausibly shift with more replications. (4) Both phases use one model, one temperature, one prompt template, and one narrow decision (single-shipment routing) – this is a finding about one policy design tested twice, not a general claim about agentic AI in logistics.

8 Follow-up: A Reinforcement-Learning Formulation

Both policies studied here are, in a precise sense, *not learned*: the heuristic is a fixed rule, and the LLM agent’s behavior comes from pre-trained general reasoning plus a prompt, not from experience inside this environment. A natural follow-up is to add a third arm that *is* learned: a reinforcement-learning (RL) policy trained directly against the simulator’s own state, action, and reward structure.

The formulation is close to already sitting in the codebase. **State:** the same read-only DecisionObservation both existing policies already receive – shipment facts, destination and shock context, current plan, candidate routes. **Action:** the same four-choice space, WAIT/REROUTE/EXPEDITE/ABSTAIN, already shared and validated identically. **Reward:** the nega-

tive of the same daily cost terms (J) already accumulated for TCD, either shaped per-day (dense) or given once at episode end as $-TCD$ (sparse, but exactly the metric this study already optimizes for by construction). Because the simulator’s transitions are already deterministic and the event tapes are already fully pre-generated and replayable, an RL agent can be trained across thousands of simulated episodes at effectively zero marginal cost – a sharply different economics from the LLM arm, where every decision is a paid, ~ 2.6 s API call, and one of the more interesting comparisons this study does not yet make: cost-per-decision at training time versus cost-per-decision at deployment time.

What it could add beyond the current two-policy comparison: **(i) an empirical reference point** – if a policy free to learn via trial and error over many episodes converges to a similar decision boundary as the heuristic or the LLM, that is independent evidence neither current policy is leaving much value on the table; if it instead discovers materially different behavior (for instance, rerouting proactively before a shock is fully disclosed, something neither current policy is built to do), that is itself a finding. **(ii) a way to separate environment from agent** in the Extended-topology reversal reported above: if a trained RL policy *also* reverses on Extended, that argues the reversal is a structural property of the decision problem itself, independent of which agent is solving it; if an RL policy does not reverse there, that argues the LLM’s specific bounded-reasoning behavior, not the environment, is responsible – a distinction Section 7 could not resolve with the two policies tested here.

This is not a free upgrade. An RL policy loses the heuristic’s auditability and the LLM’s natural-language reasoning trace – its decision boundary is a learned function, not a rule a person can read. Reward shaping is itself a research decision with failure modes (a dense per-day reward can encourage short-sighted behavior; a sparse terminal reward is harder to learn from). And a policy trained on one topology/severity distribution is not guaranteed to generalize to another without deliberately training across the full nine-cell grid, which reintroduces, at training time, exactly the topology-dependence this study’s headline result already shows matters.

9 Conclusion

Two phases, one shared paired-simulation infrastructure, and 798 live-API replications give a conditional but well-evidenced answer to whether agentic AI mitigates supply-chain disruption better than a classical heuristic: **it depends on both disruption severity and network structure, and the dependence on structure is not monotonic in the way Phase 1 alone would have suggested.** The agent’s advantage is real and large on constrained-to-moderate networks under severe disruption; the heuristic’s advantage is equally real, and grows rather than shrinks, on the most redundant network tested. Phase 1’s audit-driven redesign into Phase 2 is itself part of the finding: an experiment that cannot produce anything but unanimous agreement is not evidence of robustness, and the discipline of building an environment able to disagree with itself is what surfaced the topology-reversal result in the first

place. Near-term, future work should decouple Extended’s shock target from its topology, equalize shipment-volume variance across tiers, and extend the demand-side and supplier-side shock types already built but not yet run at grid scale; longer-term, the reinforcement-learning formulation above offers a way to test whether either policy studied here is close to a ceiling, or whether the topology-reversal finding is about the environment rather than the agent.

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